***ZCAS UNIVERSITY***

***SCHOOL OF COMPUTING, TECHNOLOGY AND APPLIED SCIENCES***

**2025 SESSIONAL EXAMINATIONS**

# HUMAN COMPUTER INTERACTION (CIT 3441)

## INSTRUCTIONS

(1) Time allowed is **Three (3) hours.**
(2) There are Seven (7) questions in this paper. Answer any Four (4) Questions.
(3) Each question carries **Twenty (25)** marks.
(4) Calculators are not allowed.
(5) The total marks for this Exam is **100**

**DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO**

**Question One [20 Marks]**

a) List several reasons why certain people resist using computers and suggest possible ways to you can alleviate these concerns. **[10 Marks]**

b) Suggest three usability measures that can be directly used to produce a practical evaluation of a system (Keep the goals of efficiency and satisfaction in mind with these measures). **[10 Marks]**

c) As noted in our lectures , some skeptics feel that accommodating diversity requires dumbing-down or lowest common-denominator strategies. However, the authors of HCI books claim that in their experience, rethinking interface designs to accommodate these diversity situations will result in a better product for all users. Give an example of a product that meets the specific needs of a certain group of people, yet gives all users a better experience. **[5 Marks]**

**Question Two**

a) The U.S. Human Engineering Design Criteria for Military Systems (1999) states four purposes of Usability requirements . State and explain these purposes. **[5 Marks]**

b) List and explain the Goals for requirements analysis. **[15 Marks]**

c) List and explain five (5) human factors central to community evaluation. **[5 Marks]**

**Question Three**

List the advantages and disadvantage of the following interaction styles: **[5 Marks Each]**

i. Direct Manipulation

ii. Menu selection

iii. Form filling

iv. Command language

v. Natural language

**Question Four**

a) State the Eight (8) golden rules of interface design. **[8 Marks]**

b) List the Seven (7) Norman 's stages of action, arranged in a cyclic pattern, as an explanatory model of human- computer interaction. **[7 Marks]**

c) Explain the three pillars of successful user-interface development. **[10 Marks]**

**Question Five**

a) As a management strategy, LUCID makes the commitment to user-centered design explicit and highlights the role of usability engineering in software development by focusing on activities, deliverables, and reviews. At each of the LUCID stages there are specified deliverables and timely feedback through reviews. List the components on which reviews are conducted. **[12 Marks]**

b) Explain the concept of Ethnographic Observation as regards to Human Computer Interface. **[13 Marks]**

**Question Six**

a) The emergence of usability testing and laboratories since the early 1980s is an indicator of the profound shift in attention to user needs. Explain the importance of usability testing and Laboratories when it comes to HCI. **[5 Marks]**

b) Discuss two products that you have interacted with, one that you find particularly easy to use and one you find difficult to use. I.e. a smartphone or a website etc. In table form answer the following questions.

1. *For the product you find easy to use, answer the following questions;* ***[10 Marks]***

    i. What is the product?

    ii. What task does the product perform?

    iii. Present ways in which this product is easy for you to use

    iv. How would someone else find this product difficult to use?

    v. How could this difficulty be resolved?

2. *For the product you find difficult to use, answer the following questions;* **[10 Marks]**

    i. What is the product?

    ii. What task does the product perform?

    iii. Present one way in which this product is difficult for you to use

    iv. How could this difficulty be resolved?

    v. How would someone else find this product easy to use?

**Questions Seven**

a) You have been asked by a software company to provide a training session on User-centered Design. Describe a user-centred design approach and techniques that you could use. **[15 Marks]**

b) Explain the goal of universality and how it can be implemented. **[5 Marks]**

c) You have been asked by a software company to provide a training session on User-centered Design. Describe a user-centred design approach and techniques that you could use. **[5 Marks]**

**END OF EXAMINATION**